

Brainstorming&Ideation Phase

Date	04 July 2026	Team ID	
Project Name	Credit Card Approval Prediction	Team Lead	Zaamir Hanan

1.1 Project Overview

Banks and financial institutions receive thousands of credit card applications every day. A significant portion are rejected due to factors such as high existing loan balances, insufficient income levels, or excessive credit inquiries. Manual review of each application is time-consuming and prone to human error. This project automates the credit card approval decision using machine learning, training a predictive model on historical applicant data to evaluate financial and demographic inputs and determine the likelihood of approval or rejection — the way real banks do. Four classification algorithms are applied: Logistic Regression, Random Forest, XGBoost (Gradient Boosting), and Decision Tree. The best-performing model is integrated into a Flask web application, with an IBM Watson Machine Learning deployment pipeline enabling scalable, real-time predictions through an intuitive interface.

1.2 Brainstorming & Idea Listing

Each team member listed ideas relevant to the problem space without judging them at this stage.

S.No	Team Member	Idea / Suggestion	Category	Group No.
1	Zaamir Hanan Mujeeb	Automate credit card approval decisions using a trained ML classifier instead of manual review	Core ML	
2	Deepak Bellam	Train and compare four models – Logistic Regression, Random Forest, XGBoost, Decision Tree – to pick the best performer	Model Building	
3	Mohana Krishna Reddy	Convert multi-class payment status codes into binary risk labels to flag high-risk applicants clearly	Feature Engineering	
4	Rushikesh Kasthuri	Build a Flask web app so analysts and applicants get an instant approval/rejection prediction	Deployment	
5	Shaik Asliman Thaslimaan	Deploy the trained model via IBM Watson ML for cloud-hosted, scalable predictions	Deployment	
6	Zaamir Hanan Mujeeb	Offer a self-service eligibility check so prospective customers can test their odds before formally applying	UX / Application	

1.3 Idea Prioritization

Each grouped idea was rated on feasibility and importance to select the final direction.

Group No.	Final Idea	Feasibility	Importance	Priority	Selected
1	ML-based automated approval prediction using four classification models	High	High	1	Yes

2	High-risk applicant flagging via engineered binary payment-status features	High	High	2	Yes
3	Flask web app + IBM Watson cloud deployment with self-service eligibility check	Medium	High	3	Yes